

10. Plasma membrane is built up of :

- (a) Protein (b) Lipid
(c) Carbohydrate (d) Both (a) and (b)

U.P.P.C.S. (Mains) 2008

Ans. (d)

The cell membrane (also known as plasma membrane) is a biological membrane that separates the interior of cell from the outside environment. It is composed of two layers of phospholipids and embedded with proteins.

11. Which of the following cell organelles play the most significant role in protein synthesis ?

- (a) Lysosome and Centrosome
(b) Endoplasmic reticulum and Ribosome
(c) Golgi apparatus and Mitochondria
(d) Lysosome and Mitochondria

I.A.S. (Pre) 2001

Ans. (b)

The rough endoplasmic reticulum (RER) is named as such because its outer membrane is littered with ribosomes. They are mainly responsible for protein synthesis.

12. The cellular and molecular control of programmed cell death are known as :

- (a) Apoptosis (b) Ageing
(c) Degeneration (d) Necrosis

I.A.S. (Pre) 2001

Ans. (a)

Apoptosis is the process of programmed cell death (PCD) that occurs in multicellular organisms and in some eukaryotic, single-celled microorganisms, such as yeast. Apoptosis plays a crucial role in developing and maintaining the health of the body by eliminating old, unnecessary and unhealthy cells.

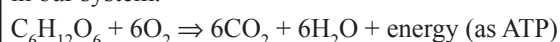
13. One term that generates the maximum number of ATP atoms in our system?

- (a) Decomposition of glucose (b) Krebs's cycle
(c) Last respiratory series (d) Water decomposition

R.A.S./R.T.S. (Pre) 1999

Ans. (b)

Krebs's cycle is a cycle of enzyme-catalyzed reactions in living cells that is the final series of reactions of aerobic metabolism of carbohydrates, proteins, and fatty acids. Due to this carbon dioxide is produced, oxygen is reduced and ATP is formed. Krebs's cycle generates the maximum number of ATP atoms in our system.



In a eukaryotic cell, the citric acid cycle occurs in the matrix of the mitochondrion. This cycle was discovered by Hans Krebs in 1937.

Heredity

Notes

- Heredity is the process of passing the traits and characteristics from parents to their offsprings.
- The offspring cells get their features, characteristics and genetic information from their mother and father.
- Heredity** and **genetics** are the reason organisms look so much like their parents.
- Genetics** is a branch of biology that studies the **genes**, **genetic variation** and **heredity** in living organisms.
- During the sexual reproduction process, genetic variation occurs due to some error in **DNA Copying**. Variation is important because it contributes to evolution and forms the basis of heredity.
- Variation is caused due to positive gene mutations, the interaction of genes with the environment and various combinations of genetic material.
- Variation can occur through asexual reproduction process too, but these variations are not very noticeable.

Gregor Johann Mendel (The Father of Genetics) :

- Acquiring characteristics or traits from one generation to the other is nothing but inheritance.
- Here, both parents contribute equally to the inheritance of traits.
- It was **Gregor Johann Mendel**, known as the **Father of Genetics**, who conducted immense research on garden pea-**Pisum sativum** and studied this inheritance of traits.
- Mendel was a **priest** in the Church of Brunn city (Austria).
- It was his research on plant breeding and hybridization that he came up with the laws of inheritance in living organisms.
- Mendel observed the pattern of inheritance from one generation to the other in pea plants which can be summarized under the following headings-
 - Law of Dominance
 - Law of Segregation
 - Law of Independent Assortment

Genome :

- A genome is an organism's complete set of DNA, including all of its genes. Each genome contains all of the information needed to build and maintain that organism. In humans, a copy of the entire genome - more than 3 billion DNA base pairs - is contained in all cells that have a nucleus.
- Gene Mapping** : Gene Mapping describes the methods used to identify the locus of a gene and the distance between genes. The essence of all genome mapping is to place a collection of molecular markers onto their respective positions on the genome.

Transposable Elements (TEs) :

- Transposable elements or transposon, also known as **jumping genes**, are DNA sequences that move from one location on the genome to others. Sometimes TEs create or reverse mutations and alter the cell's genetic identity and genome size. These elements were first identified by geneticist **Barbara McClintok** and for this discovery, she was awarded Nobel Prize in 1983.
- The production of spotted seeds in maize is due to the jumping genes.
- Almost half of our DNA sequences are made of jumping genes. They jump around the genome in developing sperm and egg cells and are important to evolution. But their mobilization can also cause new mutations that lead to diseases such as haemophilia and cancer.

Nucleic Acid :

- The term **nucleic acid** is the overall name for **DNA** (deoxyribonucleic acid) and **RNA** (ribonucleic acid). They are composed of nucleotides which are the monomers made of three components : a 5-carbon sugar, a phosphate group and a nitrogenous base.
- Nitrogenous bases are of two types- (i) Purine (ii) Pyrimidine.
- Purine bases are **adenine** and **guanine**, which are found in both DNA and RNA. Pyrimidine bases are **cytosine**, **thymine** and **uracil**. Cytosine is found in DNA and RNA both, thymine is present in DNA and uracil is present in RNA.
- The pentose sugar found in DNA is deoxyribose sugar and pentose sugar found in RNA is ribose sugar.
- Copying of DNA into RNA is called **transcription**. **Reverse transcription** is the process when the DNA copy is formed from the RNA. In reverse transcription, a reverse transcriptase (RT) enzyme takes part.
- It is noteworthy that DNA is the hereditary material in humans and almost all other organisms.
- DNA was first identified and isolated by **Friedrich Miescher** in 1869.
- Its '**Double Helix Model**' was propounded by **Watson** and **Crick** in 1953.
- DNA carries genetic instructions for the development, functioning, growth and reproduction of all known organisms and many viruses.
- It carries hereditary information from parents to offsprings.

Human Chromosomes :

- Human is unisexual animal. In human, each cell normally contains 23 pairs of chromosomes, for a total of 46.

Twenty-two of these pairs called **autosomes** in males and females alike. The 23rd pair, the sex chromosomes, differ between males and females.

- In males, the chromosomes are 44 + XY and in females, chromosomes are 44 + XX.
- The offspring which get XY chromosome from parents is male and offspring which get XX chromosome from parents is female.

Some terms about heredity

- **Gene** : It is the basic unit of inheritance. It consists of a sequence of DNA which is the genetic material. The genes can mutate and can take two or more alternative forms.
- **Chromosomes** : These are thread-like structures of nucleic acids and protein that are found in the nucleus of most living cells. They carry the heredity or genetic information in the form of genes. These are made up of nucleoprotein. German anatomist **Waldayer** is known for naming the chromosome. The number of chromosomes varies from species to species but it remains constant for a particular species. In most animals, however, any change from the typical chromosome number for a species may be due to genetic disorders.
- **Genotype** : It is the complete heritable genetic identity of an organism. It is the actual set up of alleles that are carried by the organisms.
- **Phenotype** : It is the description of the actual physical characteristics of an organism, the way the genotype is expressed.
- **Alleles** : The alternative forms of the genes which arise as a result of mutation. They are found in the same place on the chromosome and affect some characteristics or trait but in alternative forms.
- **Dominant alleles** : When an allele affects the phenotype of an organism, then it is a dominant allele. It is denoted by the capital letter. For example- 'T' to express tallness.
- **Recessive alleles** : An allele that affects the genotype in the absence of the dominant allele is called a recessive allele. It is denoted by a small letter. For example - 't' for dwarfness.
- **Homozygous** : Each organism has two alleles for every gene (each chromosome has one each). If both the alleles are same it is called homozygous. If tallness is the trait, it is expressed as 'TT'.
- **Heterozygous** : If the two alleles are different from each other, then they are heterozygous in nature. If tallness is the trait, then it is expressed as 'Tt'.

Question Bank

1. Who is the father of modern genetics?

- (a) Gregor Johann Mendel (b) Hugo De Vries
(c) Charles Darwin (d) Thomas Hunt Morgan

Chhattisgarh P.C.S. (Pre) 2011

Ans. (a)

The law of Inheritance was discovered by Gregor Johann Mendel. He is known as the father of modern genetics.

2. The scientist known as father of genetics is –

- (a) Johnson (b) G.J. Mendel
(c) F.B. Morrison (d) Morgan

Uttarakhand Lower Sub. (Pre) 2010

Ans. (b)

See the explanation of above question.

3. Who is known as father of Genetics?

- (a) Hargobind Khurana (b) Thomas Hunt Morgan
(c) Robert Brown (d) Johann Gregor Mendel

Uttarakhand P.C.S. (Pre) 2021

Ans. (d)

See the explanation of above question.

4. The classical Mendel's work on his 'laws of inheritance' was performed on the organism –

- (a) Drosophila (b) Snapdragon
(c) Garden pea (d) Sweet Pea

U.P.P.C.S. (Mains) 2002

Ans. (c)

Hybridization was experimented on garden pea plant first by Gregor Mendel. After studying the seven types of qualities of pea, he rendered three laws as- Law of Dominance, Law of Segregation and Law of Independent Assortment.

5. Mendel's principle of inheritance is based on :

- (a) Vegetative reproduction (b) Asexual reproduction
(c) Sexual reproduction (d) All of the above

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (c)

Mendel's principle of inheritance is based on sexual reproduction.

6. Genome mapping relates to :

- (a) Blood grouping
(b) Mapping of genes
(c) Mapping of nerve centers
(d) Mapping of brain

U.P.P.C.S. (Pre) 2002

Ans. (b)

Gene mapping is the method used for determining the location of gene and relative distances between genes on a chromosome. The essence of all genome mapping is to place a collection of molecular markers onto their respective positions on the genome. Molecular markers come in all forms. Genes can be viewed as one special type of genetic markers in the construction of genome maps and mapped the same way as any other markers.

7. The unit of heredity in our body is called :

- (a) Chromosome (b) DNA
(c) Gene (d) Nucleus

U.P.P.S.C. (GIC) 2010

Ans. (c)

A gene is the basic physical and functional unit of heredity. Genes, which consists of DNA, are the genetic material and unit of heredity. In humans, genes vary in size from a few hundred DNA bases to more than 2 million bases.

8. Gene is :

- (a) A segment of DNA
(b) A segment of DNA and histone
(c) A segment of DNA, RNA and histone
(d) All of the above

Uttarkhand P.C.S. (Pre) 2016

Ans. (a)

In each chromosome, DNA has a very long and coiled molecule and the small parts of this molecule are called gene. Generally a gene contains 5000 to 100000 pair nucleotide monomers of DNA molecules.

9. The characteristics features of genetic code are :

- i. It is always universal
ii. It is triplet of nucleotides bases corresponding to 20 amino acids
iii. It is non-overlapping, non-ambiguous and commaless.

iv. It has start and stop codon

Which of the statements given above are true?

- (a) i, ii and iv only (b) i, iii and iv only
(c) i, ii and iii only (d) All of the above

M.P. P.C.S. (Pre) 2020

Ans. (d)

The genetic code is the set of rules used by living cells to translate information encoded within genetic material (DNA or mRNA sequences of nucleotide triplets, or codons) into proteins. In other words, the genetic code may be defined as the exact sequence of DNA (or mRNA) nucleotides read as three letter words or codons, that determines the sequence of amino acids in protein synthesis. There are 64 codons

which correspond to 20 amino acids and to signals for the initiation and termination of transcription. Each triplet (codon) specifies one amino acid in a protein structure or a start signal or stop signal in protein synthesis. With three exceptions, each codon encodes for one of the 20 amino acids used in the synthesis of proteins. There are 61 sense codons in the genetic code which code for 20 amino acids. Those codons that code for signals during protein synthesis are known as signal (start and stop) codons. There are four codons which code for signal. These are AUG (start codon), UAA, UAG and UGA (stop codons). AUG codon also codes for the amino acid methionine. The genetic code is always universal, commaless, non-overlapping, non-ambiguous and redundant.

10. Change in the base sequence within a gene is called:

- (a) Mutation
- (b) Cloning
- (c) Fusion
- (d) Breeding

U.P.P.C.S. (Pre) 2020

Ans. (a)

A gene is the basic physical and functional unit of heredity. Change in the base sequence within a gene is called mutation. Mutation is the changing of the structure of a gene, resulting in a variant form that may be transmitted to the subsequent generations. Mutation is the ultimate source of all genetic variation, providing the raw material on which evolutionary forces such as natural selection can act.

11. Which of the following statements is not true?

- (a) Chromosomes are composed of DNA and Protein.
- (b) Nucleus of a cell does not play a role in cellular reproduction.
- (c) Chromosomes contain information for inheritance.
- (d) Functional segment of DNA is called genes.

U.P.P.C.S. (Pre) 2024

Ans. (b)

The cell nucleus is a membrane-bound organelle found in eukaryotic cells, acting as the control center and repository of genetic information. The cell nucleus, containing the cell's genetic material (DNA), plays a crucial role in cell reproduction by housing and replicating the DNA, which is then distributed to daughter cells during processes like mitosis. Hence, statement of option (b) is incorrect, while other three statements are correct.

12. Hereditary symptoms are moved in the descendant of creatures by–

- (a) Ribosome
- (b) Chromosome
- (c) Plasma
- (d) Lysosome

R.A.S./R.T.S. (Pre) 1992

Ans. (b)

In the descendant of creatures, the hereditary symptoms are moved by chromosomes. These are made from nucleoproteins. The term chromosome was coined by Waldeyer. The small pieces of chromatin during the partition of cells are known as chromosome, on which genes are found.

13. Which of the following is responsible for controlling hereditary characters of the living cells?

- (a) Enzyme
- (b) Hormone
- (c) RNA
- (d) DNA
- (e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (d)

DNA is responsible for storing and transferring genetic information while RNA directly codes for amino acids and acts as a messenger between DNA and ribosomes to make proteins. However studies showed that RNA replaces DNA in some viruses like Tobacco Mosaic virus, QB bacteriophage, etc. So, not every organism has DNA as their hereditary material, some have RNA. But, DNA is the predominant one and in the majority of higher level organisms, RNA plays the role of a messenger.

14. Which of the following features of DNA makes it uniquely suited to store and transmit genetic information from generation to generation ?

- (a) Complementarity of the two strands
- (b) Double helix
- (c) Number of base-pairs per turn
- (d) Sugar-phosphate backbone

I.A.S. (Pre) 2001

Ans. (a)

DNA is a double helix with the sugar-phosphate backbone on the outside of the molecule. The bases are on the inside, oriented such that hydrogen bonds are formed between purines and pyrimidines on opposite chains. The base pairing is very specific - A will always pair with T, G with C and the amount of adenine (A) is always equal to that of thymine (T) and the amount of guanine (G) to that of cytosine (C). Because of this specific base pairing, the two strands of a DNA molecule are complementary (each strand contains all the information required to specify the sequences of bases on the other). Complementary base pairing between DNA strands makes it uniquely suited to store and transmit genetic information from generation to generation. This complementary base pairing allows cells to copy information from one generation to another and even find and repair damage to the information stored in the sequences.

15. Sexual reproduction causes genetic variation because of

- (a) Blending of genes (b) Chromosomal changes
(c) Shuffling of genes (d) All of the above

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (d)

Sexual reproduction involves two parent cells; each parent gives some of its traits (characteristics) to the offspring. Sexual reproduction causes, genetic variation because of blending of genes, chromosomal changes, shuffling of genes etc.

16. Chromosomal theory of inheritance was given by :

- (a) Waldeyer (b) Mendel
(c) Sutton and Boveri (d) Muller

Jharkhand P.C.S. (Pre) 2021

Ans. (c)

The chromosomal theory of inheritance is credited to papers by Walter Sutton in 1902 and 1903, as well as to independent work by Theodor Boveri during roughly the same period. It is a fundamental unifying theory of genetics which identifies chromosomes as the carriers of genetic material. It correctly explains the mechanism underlying the laws of Mendelian inheritance by identifying chromosomes with the paired factors (particles) required by Mendel's laws. It also states that chromosomes are linear structures with genes located at specific sites called loci along them. Although it was Mendel who had initially put forth the theory of genes affecting character, he was unable to gain any recognition for it. Then in 1900, three scientists, de Vries, Correns and von Tschermak independently discovered chromosomes existing inside the nucleus, thus proving Mendel's theory. This theory was then firmly established in 1902-1903 by Sutton and Boveri. They compared the behaviours of chromosomes to genes and deduced certain parallels, thus justifying Mendel's laws. In 1915, Thomas Hunt Morgan's work on inheritance and genetic linkage in the fruit fly *Drosophila melanogaster* provided incontrovertible evidence for this theory. Waldeyer has coined the term "chromosome" to describe basophilic stained filaments inside the cell nucleus. It is to be noted that JPSC said in revised answer key that all options are incorrect.

17. Who first outlined the structure of the gene molecule (D.N.A.)?

- (a) Dr. Maghnad Saha
(b) Dr. Stephan Hawking
(c) Dr. James Watson and Dr. Francis Crick
(d) Dr. Alexander Fleming

M.P.P.C.S. (Pre) 1994

Ans. (c)

James Dewey Watson (born April 6, 1928) was an American molecular biologist, geneticist and zoologist, best known as one of the co-discoverers of the structure (double helix model) of DNA in 1953 with Francis Crick. Watson, Crick, and Maurice Wilkins were awarded the 1962 Nobel Prize in Physiology or Medicine for their discoveries concerning the "molecular structure of nucleic acids (DNA) and its significance for information transfer in living material".

18. James D. Watson and Francis Crick are associated with the discovery of

- (a) Vaccine (b) Structure of DNA
(c) Anti-malarial drugs (d) Penicillin

U.P.P.C.S. (Mains) 2012

Ans. (b)

See the explanation of above question.

19. The DNA double helix structure was discovered by :

- (a) James Watson and Francis Crick
(b) Rosalind Franklin and Maurice Wilkins
(c) Linus Pauling
(d) Gregor Mendel

69th B.P.S.C. (Pre) 2023

Ans. (a)

See the explanation of above question.

20. Who has proposed for the first time the double helix model of DNA ?

- (a) Watson and Crick (b) Fisher the Haldoni
(c) Lamark and Darwin (d) Hugo de Vries

U.P.P.C.S.(Pre) 2012

Ans. (a)

See the explanation of above question.

21. Nobel Prize winning scientist James D. Watson is known for his work in which area?

- (a) Metallurgy (b) Meteorology
(c) Environmental protection (d) Genetics

I.A.S. (Pre) 2008

Ans. (d)

See the explanation of above question.

22. Double-helical structure of DNA was given by whom?

- (a) Jacob and Monod (b) Watson and Crick
(c) H.G. Khorana (d) None of the above

U.P. P.C.S. (Pre) 2018

Ans. (b)

Friedrich Miescher first identified and isolated (discovered) DNA in 1869. Double helix model of DNA was given by James Watson and Francis Crick in 1953.

23. Who discovered DNA?

- (a) James Watson and Francis Crick
- (b) Gregor Mendel
- (c) Johannsen
- (d) Hargovind Khorana
- (e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (e)

See the explanation of above question.

24. Har Gobind Khorana is credited for the discovery of –

- (a) Synthesis of Protein
- (b) Synthesis of gene
- (c) Synthesis of Nitrogenous
- (d) None of these

39th B.P.S.C. (Pre) 1994

Ans. (a)

The Nobel Prize in physiology or medicine in 1968 was awarded jointly to Robert W. Holley, Har Gobind Khorana and Marshall W. Nirenberg for their interpretation of the genetic code and function of DNA in protein synthesis.

25. In laboratory, who did the synthesis of DNA –

- (a) Miller
- (b) Khorana
- (c) D. Weris
- (d) Kelvin

43rd B.P.S.C. (Pre) 1999

Ans. (b)

See the explanation of above question.

26. Who delivered the principle of 'jumping genes'?

- (a) Gregor Johann Mendel
- (b) Tomas Hunt Morgan
- (c) Barbara McClintock
- (d) Watson and Crick

I.A.S. (Pre) 1993

Ans. (c)

Barbara McClintock delivered the principle of 'jumping genes'. She was awarded Nobel Prize in 1983 for this special research in medical science while studying of heredity in the corn plant. He saw that some hereditary elements are able to move chromosomes. She found in her study that there are some spots on the grains of corn by the hereditary changes done by jumping genes. Then the conclusion is given that every cell have the part of DNA (Transposon) which are important in development.

27. Which compound present in D.N.A. does not form amino acid ?

- (a) Adenine
- (b) Tyrosine

(c) Guanine

(d) Cytosine

U.P.P.C.S. (Pre) 1994

Ans. (b)

Tyrosine or 4-hydroxyphenylalanine is one of the 22 amino acids that are used by cells to synthesize proteins. Its codons are UAC and UAU. It is a non-essential amino acid with a polar side group. It was first discovered in 1846 by German chemist Justus von Liebig in the protein casein from cheese. It is called tyrosyl when referred to as a functional group or side chain.

28. The sugar present in DNA is :

- (a) Glucose
- (b) Fructose
- (c) Deoxyribose
- (d) Ribose

U.P. P.C.S. (Mains) 2016

Ans. (c)

The deoxyribose is sugar found in DNA. It is a variant of five carbon sugar called ribose. The DNA is an informational molecule found mainly in the nucleus of the cell.

29. Ribozymes are :

- (a) DNA
- (b) RNA
- (c) Proteins
- (d) None of these

Jharkhand P.C.S. (Pre) 2016

Ans. (b)

A ribozyme is a ribonucleic acid (RNA) enzyme that catalyzes a chemical reaction. The ribozyme catalyses specific reactions in a similar way to that of protein enzymes. Also called catalytic RNA, ribozymes are found in the ribosome where they join amino acid together to form protein chains. Thomas R. Cech and Sidney Altman were awarded Nobel Prize in 1989 for discovering catalytic properties of RNA.

30. Chromosome number in any species :

- (a) Increase by age
- (b) Decrease by age
- (c) Is constant
- (d) Increase with the weight

Chhattisgarh P.C.S. (Pre) 2019

Ans. (c)

The number of chromosomes varies from species to species but it remains constant for a particular species. It does not change with the age or weight. In most animals, however, any change from the typical chromosome number for a species may be due to genetic disorders.

31. Which of the following does not have sex-chromosome?

- (a) Monkey
- (b) Tiger
- (c) Butterfly
- (d) Lizard

U.P.P.C.S. (Spl) (Mains) 2004

Ans. (d)